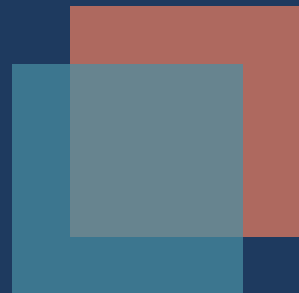


The Power of Vaccination

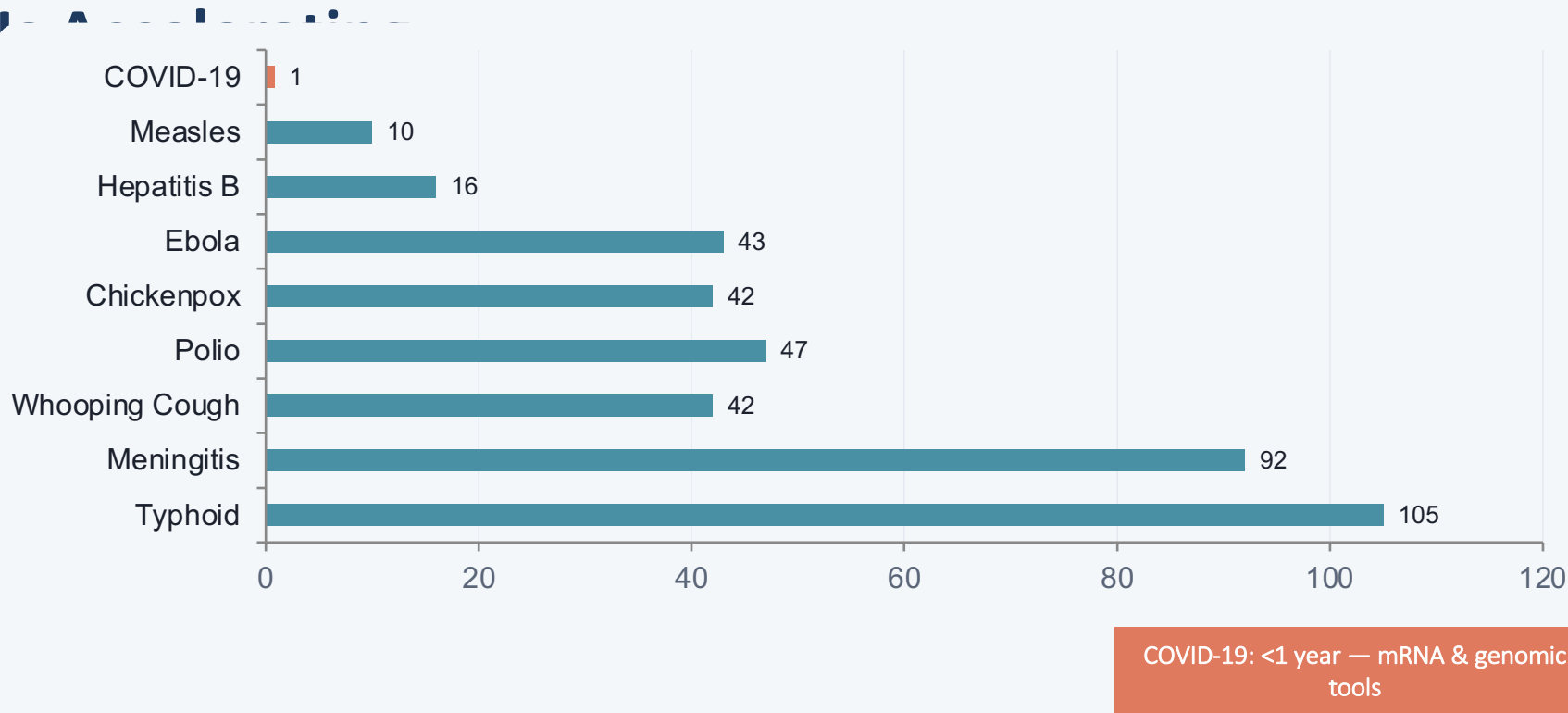
From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview: 1796 – 2020

Source: Our World in Data | Vaccination



From 100 Years to Under 1 Year: Vaccine Development



100% Case Reduction Achieved for Diphtheria, Polio,

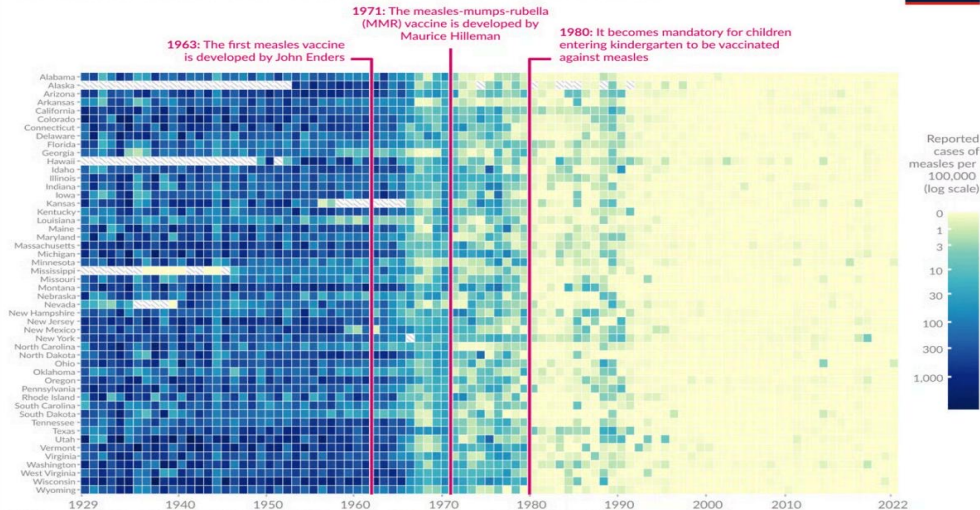
Disease	Pre-Vaccine Cases	Post-Vaccine Cases	Case Reduction	Death Reduction
Diphtheria	158	0	100%	100%
Smallpox	250	0	100%	100%
Acute polio	141	0	100%	100%
Measles	3,044	0.2	99.99%	100%
Rubella	242	0.04	99.98%	100%
Pertussis	1,534	52	96.6%	99.7%
Hepatitis A	465	51	89%	88.7%
Acute hepatitis B	273	44	83.9%	83.6%
Pneumococcal	233	139	40.5%	31.3%

■ 100% elimination
 ■ Partial reduction
 ■ Significant gaps remain

Measles Cases Collapsed Across All 50 US States After

Vaccines reduced measles cases across US states

Our World
in Data



Data source: Project Tycho (2018); Centers for Disease Control and Prevention (1959–2022)

OurWorldinData.org — Research and data to make progress against the world's largest problems.

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[Download image or data](#)

1963

First measles vaccine developed by John Enders

1971

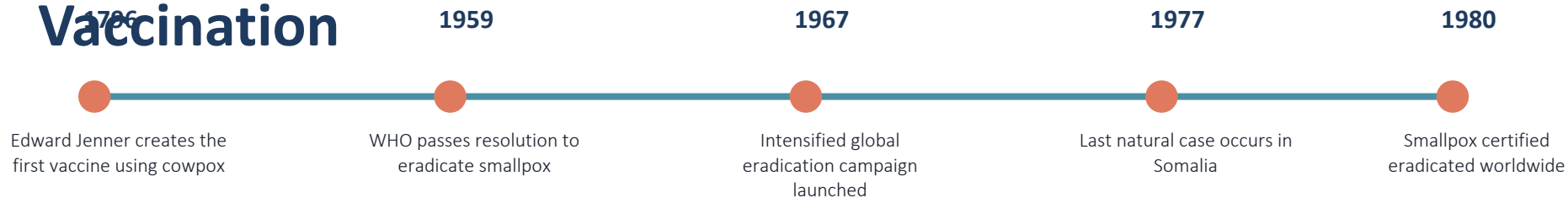
MMR (measles-mumps-rubella) vaccine by Maurice Hilleman

1980

Vaccination mandatory for kindergarten entry

Before 1963: 100–1,000+ cases per 100,000 in nearly every state

Smallpox: The Only Human Disease Eradicated Through Vaccination



Human-only host

The virus infected only humans — no animal reservoir to sustain transmission.

Visible symptoms

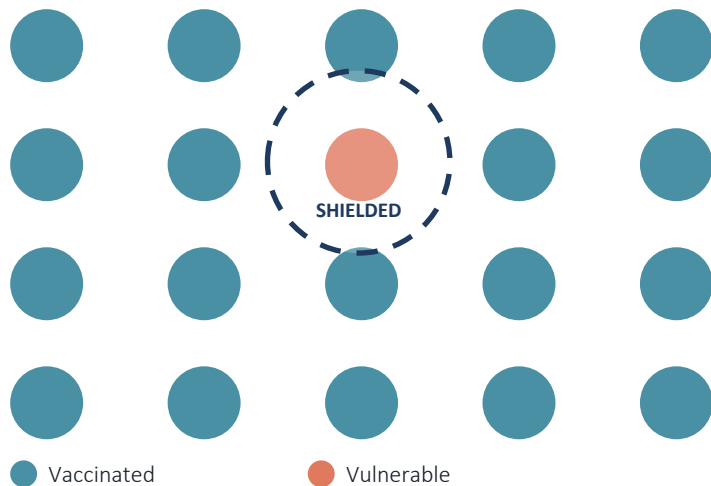
Distinctive rash made cases easy to identify and isolate quickly.

Cheap, effective vaccine

The vaccine was affordable, stable, and conferred long-lasting immunity.

Herd Immunity: How Vaccinating Many Protects the

Population-Level Protection



When enough people are vaccinated, pathogens cannot spread effectively.

- Newborns too young for vaccination
- Immunocompromised individuals
- Cancer patients on chemotherapy
- People with severe allergies to vaccine components

Gaps Persist: Pneumococcal Disease Shows Only 40%

40.5%

case reduction | 31.3% death reduction

Pneumococcal Disease

The smallest decline among major vaccine-preventable diseases in the US. Improved conjugate vaccines and broader coverage are needed.

83–89%

case reduction

Hepatitis A & B

Substantial progress, but incomplete control. Outbreaks continue where childhood immunization rates lag.

~20%

of children worldwide miss basic vaccines

Global Access & Trust

Supply shortages, geographic barriers, and declining public trust reduce uptake in vulnerable populations.

Key Takeaways

- 1 Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and both forms of polio in the US.
- 2 Vaccine development timelines have collapsed from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools.
- 3 Measles cases dropped to near zero across all 50 US states following the 1980 kindergarten vaccination mandate.
- 4 Smallpox remains the only human disease eradicated through vaccination, certified in 1980 after a global campaign.
- 5 Gaps persist for pneumococcal disease (40.5%), hepatitis A (89%), and hepatitis B (83.9%), highlighting the need for improved access, coverage, and newer vaccines.

Call to Action: Strengthen global health systems to close coverage gaps, ensure supply security, and rebuild public trust in vaccination.